UTSAV CHAUDHARY

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Oh, hello there! I am a **Fullstack Developer** with **~5 years** of experience building scalable end to end web applications and **AI driven automation systems**. Skilled in modern frontend frameworks including **React.js**, **Next.js**, **TypeScript**, and **TailwindCSS**, paired with robust backend development using **Node.js (Express)**, **Python**, and **FastAPI**. Experienced in training and integrating **AI and ML workflows** such as **LLMs**, **fine tuned models**, **speech synthesis**, and **Whisper**, enabling intelligent automation pipelines and large scale data enrichment systems. Proficient with **AWS**, **Docker**, **Kubernetes**, **CI/CD**, and distributed architectures. Adept at designing **secure**, **high performance**, **cloud native systems** with real time data processing, analytics dashboards, and multi channel automation across web and content platforms.

Technologies

Core Technologies: Python, TypeScript, JavaScript, React.js, Next.js, FastAPI, Node.js, TailwindCSS, Firebase, Redis, AWS (EC2, EKS, Lambda, RDS), Docker, Kubernetes, Redux, REST APIs, OpenAPI

Strong Working Knowledge: Django, Flask, ASP.NET, Angular, D3.js, Chart.js, WebSockets, GraphQL, WebRTC, MQTT, Azure (App Services, Azure Kubernetes Service), Kafka, Selenium, FFMPEG, OpenAI API, Ollama, Cron, RBAC, GitHub, Jira, PostgreSQL, MongoDB, Azure SQL Database, DynamoDB, AWS RDS, Terraform, Jenkins, Ansible, Nginx, Bash, PowerShell, CI/CD (GitHub Actions, Bitbucket Pipelines, AWS CodePipeline, Azure DevOps),

Past Experience / Exposure: Java, C++, C#, Embedded Systems, Raspberry Pi, Arduino, Linux (Ubuntu, CentOS), Windows, macOS, AR/VR, Game Development, IoT, Battlebots, Robotics

Experience

Fullstack Developer, Labs196, TX (Contract) Jan 2024 - Present

- Designed and built **AI agents** and **MCP servers** to extend LLM capabilities with custom tools, file systems, and APIs, enabling agents to automate codebase exploration, file operations, and task execution within the workspace.
- Trained and fine tuned **LLM-based models** for domain specific tasks; deployed inference endpoints on **AWS EKS** with **FastAPI** and integrated agents with **OpenAI** and **Ollama** for hybrid cloud and local inference.
- Automated developer and content workflows by connecting **AI agents** to **MCP servers** for search, runbooks, and tool use, reducing manual lookup time and standardizing workspace automation.
- Built agent orchestration pipelines with **Python** and **TypeScript**, integrating **React.js** and **Next.js** dashboards for monitoring agent runs, tool usage, and model performance.
- Containerized agent services and MCP servers with **Docker** on **AWS EKS**; configured **CI/CD** with **Bitbucket** and frontend on **Amplify**; collaborated with QA using **Jira**, maintaining **94% SLA resolution rate**.

Python Developer CO-OP, Midh Technologies, TX June 2023 - Dec 2023

- Fine tuned the **Llama 3.2** model on GPU backed workloads in **AWS EKS** and deployed low latency inference endpoints using **FastAPI** behind **API Gateway** with authentication and throttling.
- Built interactive analytics dashboard with **React.js** and **D3.js**, implementing granular **RBAC** for internal & client facing access.
- Integrated **Fireblocks SDK** and **Metamask** into a unified user dashboard, enabling secure on chain ID creation and blockchain asset interactions via **AWS Lambda**.
- Built and deployed the company web platform using **Next.js** with secure **RBAC** and scalable API integrations in **Amplify**.

Python/Web Developer, Compendious Medialabs Pvt. Ltd., IN June 2021 - July 2022

- Developed a responsive **React frontend** that consumed live data from the FastAPI backend, creating interactive **charts**, **graphs**, and **metrics dashboards** using **D3.js** and **Chart.js**, improving data visualization efficiency by **45%**.
- Developed an in-house **Python** scraping tool using **Selenium** and **BeautifulSoup** to automate data extraction, reducing manual collection time by **60%**.
- Implemented real-time updates using **WebSockets**, enabling continuous metric streaming for high-frequency datasets.
- Built and optimized **FastAPI** backend with structured **Pydantic** models and REST endpoints, with **250ms** avg response time.
- Improved frontend performance by refining **React state management** patterns and reducing unnecessary re-renders.

Software Engineer Remote, VRoKCs, MO March 2020 - Sept 2020

- Collaborated with ML engineers to build a full **Sign Language translation pipeline**, contributing to **data collection**, **annotation**, and **dataset preparation**.
- Built a **Flask web application** (Python, HTML5, CSS3, JavaScript) to demo and evaluate sign-language models, integrating **TensorFlow Lite** for **real-time, edge-optimized inference** directly in the browser.
- Connected the ML pipeline with **Unity3D**, enabling **real-time ASL translation** inside **Augmented Reality** environments.

- Implemented features including **image upload**, **webcam streaming**, and **live prediction display** with JS and HTML5.

Web/Game Developer, underDogs Gaming Studios, IN March 2019 - Aug 2019

- Integrated and optimized the **Quarters API** across multiple games, facilitating seamless universal **gaming currency transactions** for creating **player wallets**, conducting **secure transactions**, and converting **points to Quarters**.
- Improved real time client server communication by validating & optimizing backend interaction flows using **Python** & **C#**.
- Built **POC** modules connecting external (**payments**, **inventory**, **shipping**) to internal game, getting unified transaction layer.
- Assisted in implementing **secure transaction methods** integrating with **Unity3D**, specializing in **API best practices**, **bug fixing**, and **comprehensive unit and integration testing** for edge cases like payment failures and network drops.

Projects

LinkedIn Reverse Search Platform **Personal Project** `linkitup.thatinsaneguy.com`

- Built a **data enrichment platform** in **Python** using fine tuned **Hugging Face models** and **prompt engineered** pipelines for LLM assisted extraction of LinkedIn profiles and **company website URLs** from unstructured data, processing **1000+ records**.
- Orchestrated a **tool based pipeline** with **n8n**: **Selenium** for scraping, **SalesQL** for validation, and LLM prompts for parsing; **Redis** and **MongoDB** for caching and storage; cut repeat request time by **45%**; added analytics with **React.js** and **D3.js**.
- Designed extraction and validation **prompts** and **MCP-style tools** for profile and company URL enrichment with quality checks; integrated **OpenAI** for fallback parsing and normalization.
- Delivered **Next.js** frontend with Excel upload and real time progress; **Firebase Auth** and **Firestore**; **Docker** and **CI/CD** (GitHub Actions) for **99.5% uptime** across enrichment services and automation workflows.

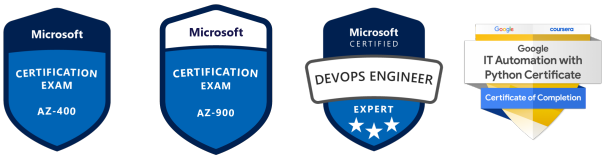
AI-Powered Video Generation Pipeline **Personal Project** `meme.thatinsaneguy.com`

- Developed an advanced **FFMPEG video generation pipeline** using **Python** and **machine learning models**, integrating **F5-TTS** for high quality voice cloning and speech synthesis to create fully AI generated videos.
- Integrated **OpenAI API** for dynamic script writing, **WhisperAI** for audio transcription, and **FFMPEG** for embedding subtitles and synchronizing audio visual elements. **Chrome** and **Selenium** for managing and automating the browser.
- Engineered a scalable video processing system that merges F5-TTS generated audio with trending gaming backgrounds, applying randomization for unique video variants using **FFMPEG** and custom overlays.
- Designed and deployed a multi channel automation system with **Docker** and **CRON jobs** for scheduled generation and publishing of **50+ videos daily** across **YouTube** and **Instagram**, including platform specific metadata and formatting.
- Built an intuitive **Gradio interface** for real time pipeline monitoring, voice cloning status updates, and batch processing.

Education

State University of New York at Binghamton 3.44/4.00 GPA
Master of Science in Computer Science

Certifications



Microsoft Certified: DevOps Engineer Expert View Credential
AZ-400: Designing and Implementing Microsoft DevOps Solutions
AZ-900: Microsoft Certified: Azure Fundamentals
Google IT Automation with Python - (Specialization) View Credential
DeepLearning.AI TensorFlow Developer - (Specialization) View Credential